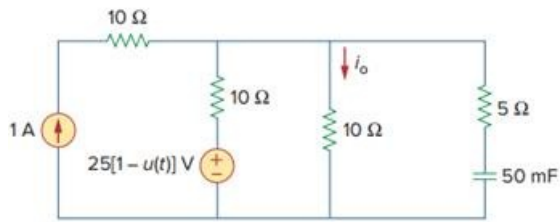


Problem

In the circuit of Fig. 7.144, find the value of i_o for all values of $0 < t$.

Fig. 7.144:



Step-by-step solution

Step 1 of 10

Refer to the Figure 7.144 in the textbook.

Consider the 1 A source for the time $t < 0$ as shown in Figure 1.

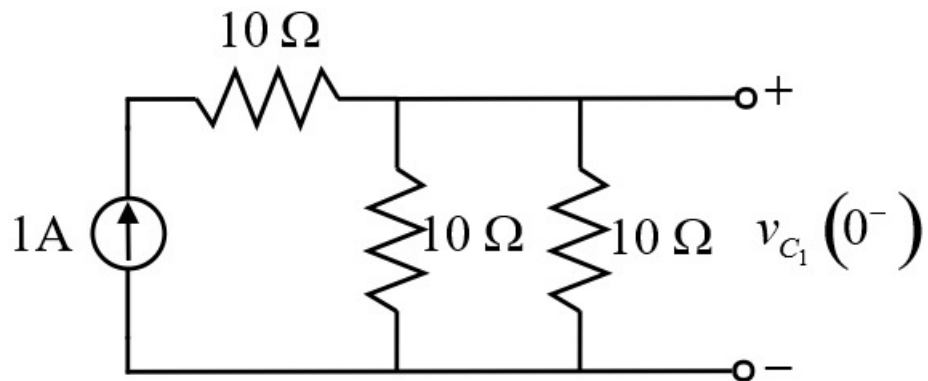


Figure 1

Step 2 of 10

Current in the $10\ \Omega$ resistor is,

$$i = 0.5\text{ A}$$

Then, the voltage $v_{C_1}(0^-)$ is,

$$v_{C_1}(0^-) = 5\text{ V} \dots\dots (1)$$

Step 3 of 10

Consider the 25 V source for the time $t < 0$ as shown in Figure 2.

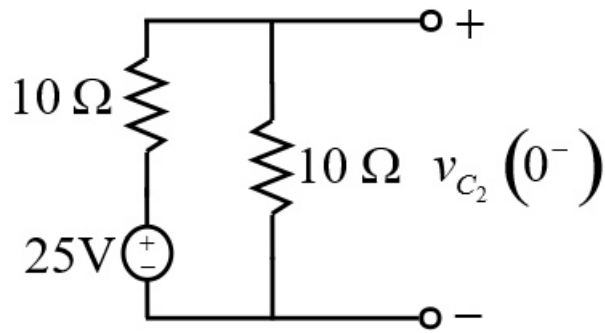


Figure 2

Step 4 of 10

Calculate the $v_{C_2}(0^-)$.

$$v_{C_2}(0^-) = \frac{25 \times 10}{20} \\ = 12.5 \text{ V}$$

Then, the voltage across the capacitor for the time $t < 0$ is,

$$v_C(0) = v_{C_1}(0^-) + v_{C_2}(0^-) \\ = 17.5 \text{ V}$$

Step 5 of 10

Refer to the Figure 7.144 in the textbook. Draw the circuit for the time $t > 0$ as shown in Figure 3.

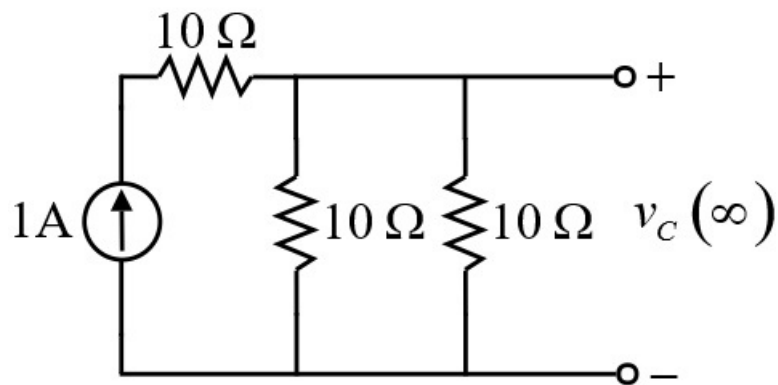


Figure 3

Step 6 of 10

Current in the $10\ \Omega$ resistor is,

$$i = 0.5\text{ A}$$

Then, the voltage $v_C(\infty)$ is,

$$v_C(\infty) = 5\text{ V}$$

Calculate the time constant τ .

$$\begin{aligned}\tau &= RC \\ &= 10 \times 50 \times 10^{-3} \\ &= 0.5\end{aligned}$$

Step 7 of 10

The voltage across the capacitor at any time t is,

$$\begin{aligned}v_C(t) &= v_C(\infty) - (v_C(\infty) - v_C(0))e^{\frac{-t}{\tau}} \\ &= 5 + 12.5e^{\frac{-t}{0.5}}\text{ V}\end{aligned}$$

Calculate the current $i_C(t)$ across the capacitor.

$$i_C(t) = C \frac{dv_C(t)}{dt}$$

Substitute 50×10^{-3} for C and $5 + 12.5e^{\frac{-t}{0.5}}$ for $v_C(t)$.

$$\begin{aligned}i_C(t) &= 50 \times 10^{-3} \frac{d\left(5 + 12.5e^{\frac{-t}{0.5}}\right)}{dt} \\ &= -1.25e^{\frac{-t}{0.5}}\text{ A}\end{aligned}$$

Step 8 of 10

Calculate the voltage across the resistor.

$$v_R(t) = 5[i_C(t)]$$

Substitute $-1.25e^{\frac{-t}{0.5}}$ for $i_C(t)$.

$$\begin{aligned}v_R(t) &= 5\left[-1.25e^{\frac{-t}{0.5}}\right] \\ &= -6.25e^{\frac{-t}{0.5}}\text{ V}\end{aligned}$$

Step 9 of 10

Then, the total voltage across resistor and capacitor is,

$$v(t) = v_R(t) + v_C(t)$$

Substitute $-6.25e^{\frac{-t}{0.5}}$ for $v_R(t)$ and $5 + 12.5e^{\frac{-t}{0.5}}$ for $v_C(t)$.

$$\begin{aligned}v(t) &= -6.25e^{\frac{-t}{0.5}} + 5 + 12.5e^{\frac{-t}{0.5}} \\ &= 5 + 6.25e^{\frac{-t}{0.5}}\text{ V}\end{aligned}$$

Step 10 of 10

Calculate the current $i_0(t)$ across the $10\ \Omega$ resistor.

$$i_0(t) = \frac{v(t)}{R}$$

Substitute $5 + 6.25e^{\frac{-t}{0.5}}$ for $v(t)$ and $10\ \Omega$ for R .

$$\begin{aligned} i_0(t) &= \frac{5 + 6.25e^{\frac{-t}{0.5}}}{10} \\ &= 0.5 + 0.625e^{\frac{-t}{0.5}}\ \text{A} \end{aligned}$$

Hence, the current across the resistor $10\ \Omega$ is $\boxed{0.5 + 0.625e^{\frac{-t}{0.5}}\ \text{A}}$.